

PHYS 143 – Mechanics and Thermal Physics

Temperature and Heat

What is temperature, how do we measure it, and how does thermal energy move?

Hope Donglo

Office 24, Department of Physics, University of Ghana

Reading: Young and Freedman, *University Physics* (15th ed.), Ch. 17, §17.1–17.4 and §17.6

By the end of this week, you should be able to:

1. Distinguish temperature from heat, and explain thermal equilibrium and the zeroth law.
2. Convert between the Celsius, Fahrenheit and Kelvin scales and describe how thermometers work.
3. Calculate thermal expansion and the heat needed to change a temperature, including simple calorimetry.
4. Calculate the rate of heat transfer by conduction and radiation, and describe convection.

The detailed outcomes are in the checklist at the end of the lecture.

Three contacts:

1 DEFINE

What is temperature, and how do we measure it?

2 MODEL

What does a temperature change do, and what does it cost?

3 APPLY

How does heat move?

Where we start: a five-minute check

Not marked. Answer individually, then we review.

1. Convert 2.5 kW to watts and 3.0 cm^2 to m^2 .
2. How much energy does a 100 W lamp use in 2.0 minutes?
3. A 2.0 m steel bar grows by 0.0010% . Find its length increase.
4. Solve $5(T - 32)/9 = 37$ for T .
5. Find the volume of a cube of side 0.10 m , in m^3 .
6. Two bodies exchange energy. If one gains 300 J , how much does the other lose?

1. 2500 W ; $3.0 \text{ cm}^2 = 3.0 \times 10^{-4} \text{ m}^2$.
2. $E = Pt = (100)(120) = 1.2 \times 10^4 \text{ J}$.
3. $\Delta L = 2.0 \times 10^{-5} \times 2.0 = 2.0 \times 10^{-5} \text{ m}$.
4. $T - 32 = 66.6 \Rightarrow T = 98.6$.
5. $(0.10)^3 = 1.0 \times 10^{-3} \text{ m}^3$.
6. It loses 300 J (energy is conserved).

Units and powers of ten (Lecture 1), energy and power (Lectures 4 and 5) are reused. Thermal physics adds one new quantity, temperature, and one new way energy is exchanged, heat.

Contact 1: Define

What is temperature, and how do we measure it?

Poll

A metal spoon and a wooden spoon have been in the same kitchen all day. You touch both. The metal one feels colder. **Which spoon has the lower temperature?**

Predict first

(a) the metal one, (b) the wooden one, (c) both are at the same temperature. What does your hand actually sense: the temperature of the spoon, or how fast energy leaves your hand?

Poll

A metal spoon and a wooden spoon have been in the same kitchen all day. You touch both. The metal one feels colder. **Which spoon has the lower temperature?**

Predict first

(a) the metal one, (b) the wooden one, (c) both are at the same temperature. What does your hand actually sense: the temperature of the spoon, or how fast energy leaves your hand?

(c). After a day in the same room both are in equilibrium with the air. Metal *conducts* energy away from your hand faster, so it *feels* colder. We will return to this in Contact 3.

- A **thermodynamic system** is the part of the universe we study; the rest is the **surroundings**.
- Its **state** is described by macroscopic variables: volume, pressure, mass, temperature.
- **Intensive** variables do not depend on the amount (temperature, pressure); **extensive** ones do (mass, volume).
- A **process** takes the system from one state to another.

Physical meaning

We describe the system with a few measurable, large-scale quantities, without tracking individual molecules. Temperature is one of these.

Predict first

Mass and temperature of a water sample: which is intensive? What happens to each if you split the sample in two?

Thermal equilibrium and walls



diathermal wall:
energy can pass



adiabatic wall:
no energy passes

Rule: thermal equilibrium

Two systems separated by a **diathermal** wall exchange energy until there is no *net* exchange: they are in **thermal equilibrium** and have the same temperature. An **adiabatic** wall prevents exchange.

Temperature is not heat

Temperature T

A property of the *state* of a body. It tells which way energy will flow when two bodies touch: from the higher to the lower temperature.

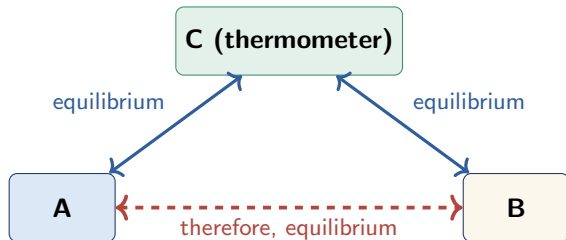
Heat Q

Energy *transferred* between bodies because of a temperature difference.
Unit: joule.

Common errors

- (1) “The body contains heat.” A body has thermal (internal) energy; heat is the energy *in transit*.
- (2) “Cold flows in.” Energy flows out of the warmer body.
- (3) “Equilibrium means no exchange.” It means no *net* exchange.

The zeroth law



Rule: the zeroth law

If A and B are each in thermal equilibrium with C , then A and B are in thermal equilibrium with each other.

Physical meaning

This is why thermometers work: C is the thermometer. If it reads the same when touching A and B , then A and B have the same temperature, even if they never touch.

How a thermometer works

- Choose a **thermometric property** that changes reproducibly with temperature.
- **Calibrate** it at fixed points (for example, melting ice and boiling water).
- Read the value when the thermometer is in equilibrium with the body.

Examples of thermometric properties:

- liquid in glass: length of the column;
- resistance thermometer: electrical resistance;
- constant-volume gas: gas pressure;
- infrared (temporal artery): radiated power.

Common error

A thermometer reads *its own* temperature. It is only meaningful once it has reached equilibrium with the body.

Temperature scales

Celsius	Fahrenheit	Kelvin	
100	-212	-373.15	steam point
37	-98.6	-310.15	body
0	-32	-273.15	ice point

Rule: scale conversions

$$T_F = \frac{9}{5}T_C + 32 \quad T_C = \frac{5}{9}(T_F - 32)$$

$$T_K = T_C + 273.15$$

Absolute zero, $0\text{ K} = -273.15^\circ\text{C}$, is the lower limit: a constant-volume gas thermometer extrapolates to zero pressure there.

TOOLKIT 1 – Temperature scales and conversions

1. **IDENTIFY:** is the quantity a *temperature* T or a temperature *difference* ΔT ?
 2. **PREDICT** the size: a body at 37°C is about 98°F and 310 K .
 3. **CONVERT a temperature:** use the formulas above, in the right order.
 4. **CONVERT a difference:** $\Delta T_K = \Delta T_C$ and $\Delta T_F = \frac{9}{5}\Delta T_C$; *no* offsets of 32 or 273.15.
- RULE:** use kelvin in any formula with T^4 or a ratio of temperatures; $^{\circ}\text{C}$ is fine for ΔT .
- WATCH OUT:** adding 273.15 to a difference, or 32 to a Fahrenheit difference.
- CHECK:** is $T_K > 0$, and does the size agree with your prediction?

Question

Normal body temperature is 37.0°C . Express it in degrees Fahrenheit and in kelvin. A fever raises it to 39.0°C : express the *change* in K, in $^{\circ}\text{C}$ and in $^{\circ}\text{F}$.

Predict first

Is the change in kelvin larger, smaller or equal to the change in $^{\circ}\text{C}$? And in $^{\circ}\text{F}$?

Question

Normal body temperature is 37.0°C . Express it in degrees Fahrenheit and in kelvin. A fever raises it to 39.0°C : express the *change* in K, in $^{\circ}\text{C}$ and in $^{\circ}\text{F}$.

Predict first

Is the change in kelvin larger, smaller or equal to the change in $^{\circ}\text{C}$? And in $^{\circ}\text{F}$?

Temperature: $T_F = \frac{9}{5}(37.0) + 32 = 98.6^{\circ}\text{F}$; $T_K = 37.0 + 273.15 = 310.15\text{ K}$.

Change: $\Delta T = 2.0^{\circ}\text{C} = 2.0\text{ K}$ (equal, no offset); $\Delta T_F = \frac{9}{5}(2.0) = 3.6^{\circ}\text{F}$.

Evaluate: equal in K and $^{\circ}\text{C}$, larger in $^{\circ}\text{F}$, as predicted. A doctor who says “310 kelvin” is not alarming. ✓

Your turn

- (a) An oven is set to 180°C . Express this in $^{\circ}\text{F}$ and in K.
- (b) Overnight the temperature falls from 12°C to -3°C . Find the change in K and in $^{\circ}\text{F}$.

Predict first

Is 180°C larger or smaller than 180°F ? Is ΔT in (b) positive or negative?

Your turn

- (a) An oven is set to 180°C . Express this in $^{\circ}\text{F}$ and in K.
- (b) Overnight the temperature falls from 12°C to -3°C . Find the change in K and in $^{\circ}\text{F}$.

Predict first

Is 180°C larger or smaller than 180°F ? Is ΔT in (b) positive or negative?

(a) $T_F = \frac{9}{5}(180) + 32 = 356^{\circ}\text{F}$; $T_K = 180 + 273.15 = 453\text{ K}$.

(b) $\Delta T = -3 - 12 = -15^{\circ}\text{C} = -15\text{ K}$; $\Delta T_F = \frac{9}{5}(-15) = -27^{\circ}\text{F}$.

Evaluate: $180^{\circ}\text{C} > 180^{\circ}\text{F}$ (since $180^{\circ}\text{F} \approx 82^{\circ}\text{C}$); the change is negative, as predicted. ✓

Contact 1: what we can now do

System and state



Thermal equilibrium, zeroth law



Thermometers and scales

You should be able to

- distinguish temperature from heat;
- state the zeroth law and explain a thermometer;
- convert temperatures and temperature differences;
- explain absolute zero.

✓ Before Contact 2

What happens to objects when their temperature changes? Read Young and Freedman, §17.3–17.4.

Contact 2: Model

What does a temperature change do, and what does it cost?

An opening question: gaps in the railway

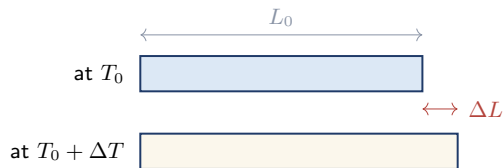
Poll

Railway tracks and bridges have small gaps (expansion joints). What is the gap for, and how large must it be for a 12 m steel rail whose temperature varies by 40 K?

Predict first

Order of magnitude: 0.1 mm, 5 mm or 50 mm?

Thermal expansion



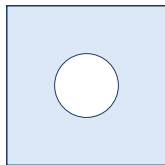
Rule: linear expansion

$$\Delta L = \alpha L_0 \Delta T$$

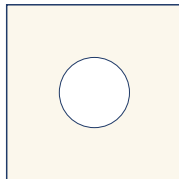
α : coefficient of linear expansion
(unit K^{-1}).

- Area: $\Delta A \approx 2\alpha A_0 \Delta T$. Volume: $\Delta V = \beta V_0 \Delta T$ with $\beta \approx 3\alpha$ for a solid.
- Valid for **small** ΔT and a uniform material; α depends slightly on temperature.

Does a hole expand or shrink?



cold plate

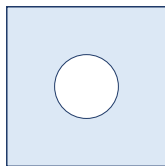


heated plate: hole grows too

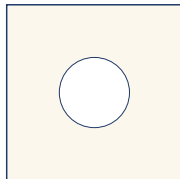
Predict first

A metal plate with a circular hole is heated. Does the hole get smaller, larger, or stay the same?

Does a hole expand or shrink?



cold plate



heated plate: hole grows too

Predict first

A metal plate with a circular hole is heated. Does the hole get smaller, larger, or stay the same?

Larger. Every length in the plate, including the diameter of the hole, scales by the same factor: the hole expands as if it were made of the same material. **Common error:** “the metal grows into the hole”.

TOOLKIT 2 – Thermal expansion

1. **IDENTIFY** the quantity that changes (length, area, volume) and the material (α from a table).
2. ΔT is final minus initial; the sign gives expansion (+) or contraction (–).
3. **CALCULATE:** $\Delta L = \alpha L_0 \Delta T$ (or $\Delta V = \beta V_0 \Delta T$, $\beta \approx 3\alpha$).
4. **COMPARE** with the size of the object (is $\Delta L/L_0$ small?).
5. **THERMAL STRESS:** if expansion is prevented, the stress is $F/A = Y\alpha\Delta T$ (Y = Young's modulus).

WATCH OUT: use L_0 at the starting temperature; hole dimensions scale like the solid.

CHECK: is the fractional change tiny (of order 10^{-4})?

Question

A steel rail is 12.0 m long at 5°C. Find its length at 45°C ($\alpha = 1.2 \times 10^{-5} \text{ K}^{-1}$). What stress would develop if the rail could not expand ($Y = 2.0 \times 10^{11} \text{ Pa}$)?

Predict first

Expansion of the order of millimetres or centimetres?

Question

A steel rail is 12.0 m long at 5°C. Find its length at 45°C ($\alpha = 1.2 \times 10^{-5} \text{ K}^{-1}$). What stress would develop if the rail could not expand ($Y = 2.0 \times 10^{11} \text{ Pa}$)?

Predict first

Expansion of the order of millimetres or centimetres?

Set up: $\Delta T = 45 - 5 = 40 \text{ K}$.

Execute: $\Delta L = \alpha L_0 \Delta T = (1.2 \times 10^{-5})(12.0)(40) = 5.8 \times 10^{-3} \text{ m} = 5.8 \text{ mm}$.

Stress: $Y \alpha \Delta T = (2.0 \times 10^{11})(1.2 \times 10^{-5})(40) = 9.6 \times 10^7 \text{ Pa}$.

Evaluate: millimetres, as predicted ($\Delta L/L_0 = 4.8 \times 10^{-4}$, tiny); the stress is large, which is why joints are needed. ✓

Your turn

A steel tank is filled to the brim with 50.0 L of gasoline ($\beta_{\text{gas}} = 9.5 \times 10^{-4} \text{ K}^{-1}$) at 10°C . The tank steel has $\beta_{\text{steel}} = 3.6 \times 10^{-5} \text{ K}^{-1}$. Estimate the volume that spills when it warms to 30°C .

Predict first

Which expands more, the gasoline or the tank? Does the spilled volume equal the gasoline's expansion alone?

Your turn

A steel tank is filled to the brim with 50.0 L of gasoline ($\beta_{\text{gas}} = 9.5 \times 10^{-4} \text{ K}^{-1}$) at 10°C . The tank steel has $\beta_{\text{steel}} = 3.6 \times 10^{-5} \text{ K}^{-1}$. Estimate the volume that spills when it warms to 30°C .

Predict first

Which expands more, the gasoline or the tank? Does the spilled volume equal the gasoline's expansion alone?

$\Delta T = 20 \text{ K}$. Gasoline: $9.5 \times 10^{-4} \times 50.0 \times 20 = 0.95 \text{ L}$. Tank:

$3.6 \times 10^{-5} \times 50.0 \times 20 = 0.036 \text{ L}$.

Spill = $0.95 - 0.04 = 0.91 \text{ L}$. **Evaluate:** gasoline expands far more than the tank, as predicted. ✓

Rule: heat and specific heat

$$Q = mc \Delta T \quad \text{unit of } c : \text{ J/(kg K)}$$

$Q > 0$: energy transferred *into* the body (its temperature rises). $Q < 0$: energy transferred *out*.

- Specific heat c : the energy per kilogram per kelvin; water: 4186 J/(kg K) .
- Heat capacity of an object: $C_{\text{obj}} = mc$.
- Molar heat capacity: $C = Mc$ (unit J/(mol K)).

Common error

A large c does not make something heat up quickly; it means a *lot* of energy is needed per kelvin. Water heats slowly.

TOOLKIT 3 – Heat and specific heat

1. **IDENTIFY** the body, its mass m and material (c from a table).
 2. **PREDICT** the sign of Q : heating $\Rightarrow Q > 0$; cooling $\Rightarrow Q < 0$.
 3. $\Delta T = T_f - T_i$ (kelvin or $^{\circ}\text{C}$, the same size).
 4. **CALCULATE** $Q = mc\Delta T$ with m in kg and c in $\text{J}/(\text{kg K})$.
 5. **USE** $Q = C_{\text{obj}}\Delta T$ for an object, or $Q = nC\Delta T$ with the molar heat capacity.
- WATCH OUT:** this formula holds only if there is no phase change; c is assumed constant over the range.
- CHECK:** does the sign of Q agree with heating or cooling?

Question

A kettle heats 1.50 kg of water from 20°C to 100°C. (a) How much heat is needed? (b) How long does this take at a heating rate of 2.0 kW (all energy going into the water)?

Predict first

Order of magnitude of Q : 10^2 , 10^5 or 10^8 J? Minutes or hours?

Question

A kettle heats 1.50 kg of water from 20°C to 100°C. (a) How much heat is needed? (b) How long does this take at a heating rate of 2.0 kW (all energy going into the water)?

Predict first

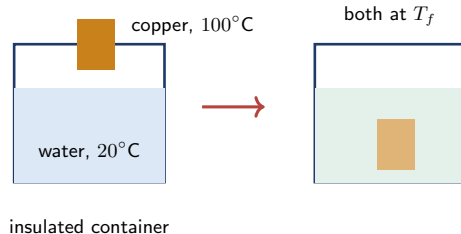
Order of magnitude of Q : 10^2 , 10^5 or 10^8 J? Minutes or hours?

$$(a) Q = mc\Delta T = (1.50)(4186)(80) = 5.0 \times 10^5 \text{ J.} \quad (b)$$

$$t = Q/P = 5.0 \times 10^5 / 2000 = 2.5 \times 10^2 \text{ s} \approx 4.2 \text{ min.}$$

Evaluate: about 10^5 J and a few minutes, as predicted; $Q > 0$. ✓

Calorimetry: energy conservation for heat



Rule: calorimetry

In an isolated system, the heat gained by the cooler parts equals the heat lost by the hotter parts:

$$\sum Q = 0 \quad \text{or} \quad Q_{\text{gained}} = -Q_{\text{lost}}$$

✂ TOOLKIT 4 – Calorimetry (no phase change)

1. **SYSTEM:** list every body that exchanges energy; insulate the system from the surroundings.
 2. **UNKNOWN** final temperature T_f (common to all bodies at equilibrium).
 3. **WRITE** $Q_i = m_i c_i (T_f - T_{i,\text{initial}})$ for each body, with signs.
 4. **SET** $\sum Q_i = 0$ and solve for T_f (a linear equation).
 5. **EVALUATE:** T_f must lie between the lowest and highest initial temperatures.
- WATCH OUT:** if T_f would reach a melting or boiling point, a phase change occurs: use the methods of the next lecture.
- CHECK:** does the heat lost by the hot body equal the heat gained by the cold one?

Question

A 0.200 kg copper block ($c = 390 \text{ J}/(\text{kg K})$) at 100.0°C is dropped into 0.500 kg of water ($c = 4186 \text{ J}/(\text{kg K})$) at 20.0°C in an insulated container. Find the final temperature.

Predict first

Will T_f be closer to 100°C or to 20°C ? Why? (Compare the heat capacities.)

Question

A 0.200 kg copper block ($c = 390 \text{ J}/(\text{kg K})$) at 100.0°C is dropped into 0.500 kg of water ($c = 4186 \text{ J}/(\text{kg K})$) at 20.0°C in an insulated container. Find the final temperature.

Predict first

Will T_f be closer to 100°C or to 20°C ? Why? (Compare the heat capacities.)

Execute: $m_{\text{Cu}}c_{\text{Cu}}(T_f - 100) + m_w c_w(T_f - 20) = 0$

$$(78.0)(T_f - 100) + (2093)(T_f - 20) = 0 \Rightarrow T_f = \frac{7800 + 41\,860}{78.0 + 2093} = 22.9^\circ\text{C}.$$

Evaluate: close to 20°C because the water's heat capacity (2093 J/K) is much larger than the copper's (78 J/K). Check: copper loses $78(77.1) = 6.0 \text{ kJ}$, water gains $2093(2.9) = 6.0 \text{ kJ}$. ✓

Your turn

0.30 kg of water at 80°C is poured into 0.50 kg of water at 20°C in an insulated container. Find the final temperature.

Predict first

Will T_f be above or below 50°C (the simple average)? Why?

Your turn

0.30 kg of water at 80°C is poured into 0.50 kg of water at 20°C in an insulated container. Find the final temperature.

Predict first

Will T_f be above or below 50°C (the simple average)? Why?

$$0.30 c (T_f - 80) + 0.50 c (T_f - 20) = 0 \Rightarrow T_f = \frac{0.30(80) + 0.50(20)}{0.80} = 42.5^\circ\text{C}.$$

Evaluate: below 50°C, as predicted, because there is more cold water than hot; between 20 and 80°C. ✓

Contact 2: what we can now do

Temperature change



expansion: $\Delta L = \alpha L_0 \Delta T$

heat: $Q = mc\Delta T$



calorimetry: $\sum Q = 0$

You should be able to

- calculate linear, area and volume expansion;
- find the heat for a temperature change, with the correct sign;
- solve a calorimetry problem with no phase change.

✓ Before Contact 3

How does the energy actually get from one body to another? Read Young and Freedman, §17.6.

Contact 3: Apply

How does heat move?

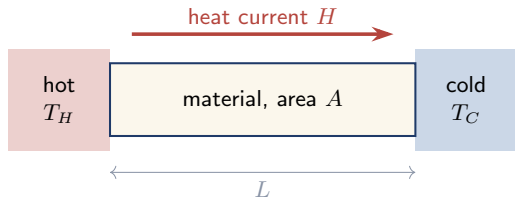
Three ways heat moves

Mechanism	How energy moves	Needs matter?	Example
conduction	collisions and vibrations between neighbouring particles, no bulk motion	yes (a solid, liquid or gas)	a spoon in hot tea
convection	the motion of the fluid itself carries the energy	yes (a fluid)	boiling water, sea breeze
radiation	electromagnetic waves	no	the Sun, a fire

Predict first

Which mechanism brings the Sun's energy to the Earth? Which warms your hand when you hold it beside a fire, not above it?

Conduction



Rule: heat current in conduction

$$H = \frac{dQ}{dt} = kA \frac{T_H - T_C}{L}$$

k : thermal conductivity (W/(m K)).
 H in watts.

- Larger A or larger ΔT gives more; a longer path gives less.
- Metals have large k ; air, foam and wood have small k (insulators). Insulators *reduce* the rate; they do not stop the flow.

TOOLKIT 5 – Conduction

1. **SKETCH** the hot side, the cold side, the path length L and the area A perpendicular to the flow.

2. **IDENTIFY** k and $\Delta T = T_H - T_C$ (in K or $^{\circ}\text{C}$).

3. **CALCULATE** $H = kA\Delta T/L$ (watts).

4. **SERIES:** when two layers carry the same current, set $H_1 = H_2$ and solve for the interface temperature.

WATCH OUT: L is the length *along* the flow, and A is the cross-section across it; the current is the same through every layer in steady state.

CHECK: is H positive from hot to cold, and is the interface temperature between T_C and T_H ?

Question

The foam lid of a cooler has area 0.60 m^2 and thickness 2.0 cm ($k = 0.033 \text{ W}/(\text{m K})$). The inside is at 5°C and the outside at 25°C . Find the heat current, and the energy conducted in one hour.

Predict first

Is the current of order 1 W , 20 W or 2000 W ?

Question

The foam lid of a cooler has area 0.60 m^2 and thickness 2.0 cm ($k = 0.033 \text{ W}/(\text{m K})$). The inside is at 5°C and the outside at 25°C . Find the heat current, and the energy conducted in one hour.

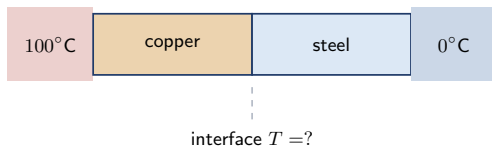
Predict first

Is the current of order 1 W, 20 W or 2000 W?

$$H = kA \frac{\Delta T}{L} = (0.033)(0.60) \frac{20}{0.020} = 20 \text{ W}. \quad Q = Ht = (20)(3600) = 7.1 \times 10^4 \text{ J per hour}.$$

Evaluate: about 20 W, as predicted: small, because k is small. This energy would warm 3.0 kg of drink by $\Delta T = Q/(mc) = 5.7 \text{ K}$ each hour. ✓

Lecturer example: two bars in series



A 0.30 m copper bar ($k = 385$) is joined end to end to a 0.30 m steel bar ($k = 50.2$), both of area 4.0 cm^2 . The ends are at 100°C and 0°C . Find the interface temperature and the heat current.

Predict first

Which bar has the larger temperature drop, copper or steel?

Two bars in series: solution

Set up: steady state: the same current in both: $H_{\text{Cu}} = H_{\text{steel}}$, and the lengths and areas are equal.

Two bars in series: solution

Set up: steady state: the same current in both: $H_{\text{Cu}} = H_{\text{steel}}$, and the lengths and areas are equal.

Execute: $385(100 - T) = 50.2(T - 0) \Rightarrow 38\,500 = 435.2T \Rightarrow T = 88.5^\circ\text{C}.$

$$H = \frac{kA(T - 0)}{L} = \frac{(50.2)(4.0 \times 10^{-4})(88.5)}{0.30} = 5.9 \text{ W}.$$

Two bars in series: solution

Set up: steady state: the same current in both: $H_{\text{Cu}} = H_{\text{steel}}$, and the lengths and areas are equal.

Execute: $385(100 - T) = 50.2(T - 0) \Rightarrow 38\,500 = 435.2T \Rightarrow T = 88.5^\circ\text{C}$.

$$H = \frac{kA(T - 0)}{L} = \frac{(50.2)(4.0 \times 10^{-4})(88.5)}{0.30} = 5.9 \text{ W}.$$

Evaluate: the interface is close to the copper's hot end: nearly all the temperature drop (88.5 K) is across the poor conductor (steel), as predicted. Check with copper:
 $(385)(4.0 \times 10^{-4})(11.5)/0.30 = 5.9 \text{ W}$. ✓

Concept check: why does metal feel colder?

Question

Back to the spoons: both are at 20°C , but your hand is at 33°C . Explain why the metal one feels colder.

Predict first

Which factor in $H = kA\Delta T/L$ differs between metal and wood?

Concept check: why does metal feel colder?

Question

Back to the spoons: both are at 20°C , but your hand is at 33°C . Explain why the metal one feels colder.

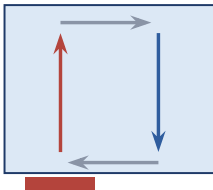
Predict first

Which factor in $H = kA\Delta T/L$ differs between metal and wood?

ΔT is the same for both, but $k_{\text{metal}} \gg k_{\text{wood}}$. The metal carries energy away from your skin much faster, so the skin cools more and the nerves report “colder”. The sensation measures the *rate* of energy transfer, not the temperature.

Convection

warm fluid rises, cool fluid sinks

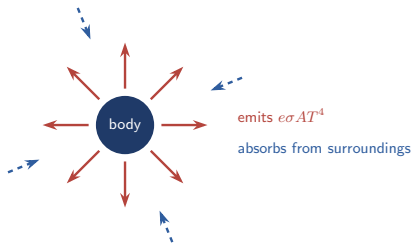


heat source below

Convection

Energy is carried by the *motion of a fluid*: warm, less dense fluid rises; cooler fluid sinks. Natural (buoyancy) or forced (fan, pump).

There is no single formula like conduction's, because the rate depends on the fluid's motion, the geometry of the surface, and the temperature difference together, and the motion is usually turbulent. Everyday examples: sea breeze, boiling, radiators, the body's blood flow.



Rule: radiation (Stefan–Boltzmann law)

$$H = e\sigma AT^4 \quad \sigma = 5.67 \times 10^{-8} \text{ W}/(\text{m}^2 \text{ K}^4)$$

e : emissivity ($0 \leq e \leq 1$); T in **kelvin**. Net exchange with surroundings at T_s :

$$H_{\text{net}} = e\sigma A (T^4 - T_s^4).$$

Every body radiates, whatever its temperature; a body at the same temperature as its surroundings still radiates, but absorbs just as much (thermal equilibrium).

TOOLKIT 6 – Radiation

1. **IDENTIFY** the surface area A , emissivity e and the temperatures of the body and surroundings.
 2. **CONVERT** all temperatures to **kelvin** (Toolkit 1).
 3. **EMITTED POWER:** $H = e\sigma AT^4$.
 4. **NET POWER** with surroundings at T_s : $H_{\text{net}} = e\sigma A(T^4 - T_s^4)$.
- WATCH OUT:** using $^{\circ}\text{C}$ in T^4 ; forgetting the radiation absorbed from the surroundings.
- CHECK:** is $H_{\text{net}} > 0$ when the body is hotter than its surroundings?

Question

A person has skin temperature 33°C , surface area 1.8 m^2 and $e = 0.97$. (a) What power do they radiate? (b) What is the net loss in a room at 20°C ?

Predict first

Order of magnitude: 10 , 10^2 or 10^4 W ? Is the net loss larger or smaller than (a)?

Question

A person has skin temperature 33°C , surface area 1.8 m^2 and $e = 0.97$. (a) What power do they radiate? (b) What is the net loss in a room at 20°C ?

Predict first

Order of magnitude: 10 , 10^2 or 10^4 W ? Is the net loss larger or smaller than (a)?

$$T = 306\text{ K}, T_s = 293\text{ K}.$$

$$(a) H = e\sigma AT^4 = (0.97)(5.67 \times 10^{-8})(1.8)(306)^4 = 8.7 \times 10^2\text{ W}.$$

$$(b) H_{\text{net}} = (0.97)(5.67 \times 10^{-8})(1.8)[306^4 - 293^4] = 1.4 \times 10^2\text{ W}.$$

Evaluate: the net loss is much smaller, because the person also absorbs from the room, and it is of order 10^2 W , comparable with the body's metabolic power. ✓

TOOLKIT 7 – Rates and energy

1. **RATE:** identify the heat current H (W) from conduction (Toolkit 5) or radiation (Toolkit 6).

2. **ENERGY:** $Q = Ht$ for a constant current over a time t .

3. **EFFECT:** link to the body: $Q = mc\Delta T$ (Toolkit 3), no phase change.

4. **SOLVE** for the unknown (t , ΔT or H).

WATCH OUT: H changes if T changes: use the initial rate for small changes only, and say so.

CHECK: are the units consistent: $\text{W} \times \text{s} = \text{J}$?

Integrated problem: heating water through a rod

Decide what physics applies; do not look for a formula

A 0.50 kg sample of water at 20°C in an insulated container is heated by a copper rod ($k = 385 \text{ W}/(\text{m K})$, length 0.40 m, cross-section 2.0 cm^2) whose other end is in a reservoir at 100°C. Ignore losses.

- (a) What is the initial heat current?
- (b) About how long does the water take to warm by 1.0 K?
- (c) Is it reasonable to use the initial current for this estimate?

Predict first

IDENTIFY → mechanism and model → PREDICT → EXECUTE → EVALUATE.
Minutes or hours?

Identify, set up: conduction along the rod (Toolkit 5), then the effect on the water (Toolkit 3), linked by $Q = Ht$ (Toolkit 7). $A = 2.0 \times 10^{-4} \text{ m}^2$.

Identify, set up: conduction along the rod (Toolkit 5), then the effect on the water (Toolkit 3), linked by $Q = Ht$ (Toolkit 7). $A = 2.0 \times 10^{-4} \text{ m}^2$.

$$(a) \quad H = kA \frac{T_H - T_C}{L} = (385)(2.0 \times 10^{-4}) \frac{80}{0.40} = 15 \text{ W}.$$

Identify, set up: conduction along the rod (Toolkit 5), then the effect on the water (Toolkit 3), linked by $Q = Ht$ (Toolkit 7). $A = 2.0 \times 10^{-4} \text{ m}^2$.

$$\text{(a)} \quad H = kA \frac{T_H - T_C}{L} = (385)(2.0 \times 10^{-4}) \frac{80}{0.40} = 15 \text{ W}.$$

$$\text{(b)} \quad Q = mc\Delta T = (0.50)(4186)(1.0) = 2.1 \times 10^3 \text{ J};$$
$$t = Q/H = 2093/15.4 = 1.4 \times 10^2 \text{ s} \approx 2.3 \text{ min}.$$

Identify, set up: conduction along the rod (Toolkit 5), then the effect on the water (Toolkit 3), linked by $Q = Ht$ (Toolkit 7). $A = 2.0 \times 10^{-4} \text{ m}^2$.

(a) $H = kA \frac{T_H - T_C}{L} = (385)(2.0 \times 10^{-4}) \frac{80}{0.40} = 15 \text{ W}.$

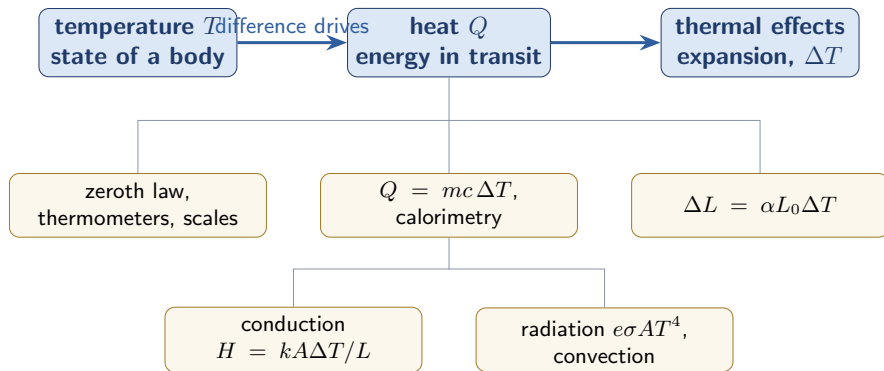
(b) $Q = mc\Delta T = (0.50)(4186)(1.0) = 2.1 \times 10^3 \text{ J};$

$t = Q/H = 2093/15.4 = 1.4 \times 10^2 \text{ s} \approx 2.3 \text{ min}.$

(c) Yes: the water warms by 1 K, so T_C changes from 20 to 21°C and ΔT from 80 to 79 K, a 1% change in H .

Evaluate: minutes, as predicted, for a small temperature change; warming all the way to 100°C would take hours, because the current falls as the water warms. ✓

Lecture 9 concept map



MASTER PROCEDURE – THERMAL

1. **IDENTIFY** The system, the bodies and the process: what is asked?
 2. **SKETCH** The bodies, the boundaries and the direction of energy flow.
 3. **UNITS** Convert temperatures: kelvin for T^4 ; differences in K or $^{\circ}\text{C}$.
 4. **PREDICT** Sign of Q , direction of heat flow, and the rough size.
 5. **SELECT THE MODEL** Scale, expansion, specific heat, calorimetry, conduction or radiation.
 6. **EXECUTE** Write the equations first, then substitute with units.
 7. **EVALUATE** Signs, units, magnitude, limiting cases, and a second check (energy conservation).
- SKETCH $\rightarrow$ PREDICT $\rightarrow$ CALCULATE $\rightarrow$ CHECK.

Inside the course framework

IDENTIFY $\leftrightarrow$ steps 1–2
SET UP $\leftrightarrow$ steps 3–5
EXECUTE $\leftrightarrow$ step 6
EVALUATE $\leftrightarrow$ step 7

Physical meaning

Next lecture: what happens when the temperature reaches a melting or boiling point (phase changes and latent heat).



THE SEVEN TOOLKITS

- | | | |
|---|------------------------------------|---|
| 1 | Temperature scales and conversions | T against ΔT ; kelvin when needed |
| 2 | Thermal expansion | $\Delta L = \alpha L_0 \Delta T$; $\beta \approx 3\alpha$; thermal stress |
| 3 | Heat and specific heat | $Q = mc\Delta T$; sign; units |
| 4 | Calorimetry (no phase change) | $\sum Q = 0$; T_f between the initial values |
| 5 | Conduction | $H = kA\Delta T/L$; series layers |
| 6 | Radiation | $H = e\sigma AT^4$; net exchange; kelvin |
| 7 | Rates and energy | $Q = Ht$ combined with $Q = mc\Delta T$ |

The Master Procedure sits above these. Earlier toolkits (units, energy and power, signs) are reused, not renamed.

Scales

$$T_F = \frac{9}{5}T_C + 32, \quad T_K = T_C + 273.15$$

Expansion

$$\Delta L = \alpha L_0 \Delta T, \quad \Delta V = \beta V_0 \Delta T, \quad \beta \approx 3\alpha$$

Heat

$$Q = mc\Delta T, \quad C = Mc$$

Calorimetry

$$\sum Q = 0 \text{ (no phase change)}$$

Conduction

$$H = kA(T_H - T_C)/L$$

Radiation

$$H = e\sigma AT^4, \quad H_{\text{net}} = e\sigma A(T^4 - T_s^4)$$

✓ Habit 1

Decide whether you have a temperature or a temperature difference.

✓ Habit 2

Use kelvin whenever T is raised to a power or divided.

✓ Habit 3

Heat is energy in transit; do not say a body contains heat.

✓ Habit 4

Predict the sign of Q and the direction of the flow first.

✓ Habit 5

Check that the equilibrium temperature lies between the initial ones.

✓ Habit 6

Convert a rate to an energy with $Q = Ht$, then use $Q = mc\Delta T$.

Checklist: can I ...

- ☐ distinguish temperature from heat?
- ☐ state the zeroth law and explain a thermometer?
- ☐ convert temperatures and temperature differences?
- ☐ calculate linear, area and volume expansion?
- ☐ calculate a thermal stress?
- ☐ find $Q = mc\Delta T$ with the correct sign?
- ☐ solve a calorimetry problem without a phase change?
- ☐ use $H = kA\Delta T/L$, including layers in series?
- ☐ use $H = e\sigma AT^4$ with kelvin?
- ☐ describe convection?
- ☐ combine a rate with a time and $Q = mc\Delta T$?
- ☐ select the appropriate model in an unfamiliar problem?

After this week

Tutorial sheet: extra conversion, expansion, specific-heat, calorimetry, conduction, radiation and combined problems. Supplementary notes: the gas thermometer and absolute zero, thermal stress, thermal resistance, and blackbody radiation in outline.